

Karan Jadhav

Senior Backend Engineer | Python, FastAPI, Django, PostgreSQL, AWS

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Summary

Senior Backend Engineer with 5+ years building production APIs, async services, database-backed systems, and AWS infrastructure. Strongest work includes FastAPI/Django REST APIs with p99 latency under 200 ms, distributed background pipelines processing 50K+ jobs/day, and PostgreSQL/PostGIS systems handling **2.3B+ records and 1.8 TB+ indexed data**.

Core Expertise

- Backend APIs: Python, FastAPI, Django, REST design, validation, pagination, async services.
- Database-Backed Systems: PostgreSQL, PostGIS, Redis, indexing, query plans, schema design.
- Async Processing: RabbitMQ, Dask, worker orchestration, retries, scheduling, failure handling.
- Performance: p99 latency, caching, query tuning, throughput improvement, database load reduction.
- Cloud Backend: AWS EC2, S3, RDS, SQS, Lambda, CloudWatch, Docker, CI/CD.
- Production Reliability: structured logging, monitoring, deployment, debugging, incident support.

Experience

Software Engineer - Intensel Ltd | 2021 - Present

- Built and owned backend systems for a production climate-risk platform serving enterprise customers across 10+ countries.
- Developed high-performance REST APIs using FastAPI and Django, supporting p99 latency under 200 ms under load.
- Designed backend services over PostgreSQL/PostGIS datasets with 2.3B+ geospatial records and 1.8 TB of indexed data.
- Reduced critical API/query latency from minutes to single-digit milliseconds through indexing, query tuning, and caching.
- Designed and operated distributed background job pipelines processing 50K+ jobs/day using RabbitMQ and Dask.
- Built fault-tolerant async execution flows with retries, scheduling, worker orchestration, and failure recovery.
- Implemented cache-aside and selective invalidation strategies to reduce database load and improve response times.
- Managed backend infrastructure on AWS using EC2, S3, RDS, SQS, Lambda, and CloudWatch.
- Built CI/CD pipelines and observability systems, reducing deployment time by 60%.
- Implemented structured logging and monitoring to improve production debugging and reliability.

Projects

Global Building Footprints

- Built backend ingestion and query infrastructure for 2.3B+ building records.
- Enabled low-latency analytics workflows by optimizing API/database access paths.

Map Tile Service

- Designed scalable backend services for geospatial raster delivery serving 5.3 TB of data.
- Improved throughput and latency through caching and performance tuning.

Open Source

RediServe | Rust, Axum, Tokio

- Built an async HTTP API for Redis with connection pooling.
- Focused on low-latency request handling and efficient backend resource usage.

Technical Skills

Languages: Python, Rust, Go, JavaScript, TypeScript, SQL
Backend: FastAPI, Django, REST APIs, async workers, API design
Databases: PostgreSQL, PostGIS, Redis, DuckDB
Cloud: AWS EC2, S3, RDS, SQS, Lambda, CloudWatch
Infrastructure: Docker, Linux, CI/CD, observability
Systems: distributed systems, message queues, caching, scalability

Education

Bachelor of Computer Science - University of Mumbai | 2020